Revisiting Continual Learning: A Three-Interface Diagnosis of Stability and Plasticity

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Abstract

Incremental learning is usually evaluated through the final classifier or LM head, but this interface does not indicate where forgetting occurs. A drop in accuracy may reflect representation degradation in the backbone, readout mismatch in the head, or both. We argue that BWT and FWT should therefore be interpreted as interface-dependent quantities rather than as single model-level properties. We study three evaluation interfaces: the sequentially trained head, a newly optimized probing classifier, and a classifier-free backbone separability diagnostic based on clustering. These interfaces answer different questions: deployed performance, recoverable information under a new readout, and representation-level separability. Across CIL experiments with encoder and decoder language backbones, plus discriminative-backbone TIL sanity checks, we find that qualitative conclusions about forgetting and plasticity change substantially with the evaluation interface. Standard heads often indicate severe forgetting, probes often suggest near-complete recoverability, and backbone diagnostics suggest intermediate representation changes and a stability–plasticity pattern. This pattern is consistent across five clustering algorithms and three clustering metrics, suggesting that it is not an artifact of a single diagnostic choice. Finally, we use Just LM-Head Tuning (JLT) as a head-only intervention to quantify recoverable performance on a fixed incrementally trained backbone. The large gap between standard and head-realigned performance suggests that many failures attributed to catastrophic forgetting are better understood as head–backbone mismatch plus partial representation drift. Our results call for reporting where forgetting is measured, not only how much forgetting is observed.

1 Introduction

Incremental Learning (IL) studies how a model can learn a sequence of tasks while retaining useful knowledge from earlier tasks [Polikar et al., 2001, Kemker and Kanan, 2017, Parisi et al., 2019]. In language modeling and NLP, this problem is especially important because pretrained language models (PLMs) are often adapted to new domains or task streams through sequential fine-tuning. A persistent challenge is catastrophic forgetting: performance on earlier tasks can degrade sharply after the model is updated on later tasks [French, 1999, Li et al., 2019, Hu et al., 2019, Kaushik et al., 2021, van de Ven et al., 2024]. Many IL methods therefore focus on memory stability through replay, regularization, or distillation [Kirkpatrick et al., 2017, Wang et al., 2023, Yang et al., 2024b].

However, standard IL evaluation usually observes the model through the final classifier or LM head. This interface is useful for measuring deployed task accuracy, but it can also mix together two different phenomena: degradation of the backbone representation and failure of the prediction head to read out information that is still present in the backbone. This distinction matters for the *Stability–Plasticity Dilemma* [Abraham and Robins, 2005, Mermillod et al., 2013, Wu et al., 2021, Araujo et al., 2022].

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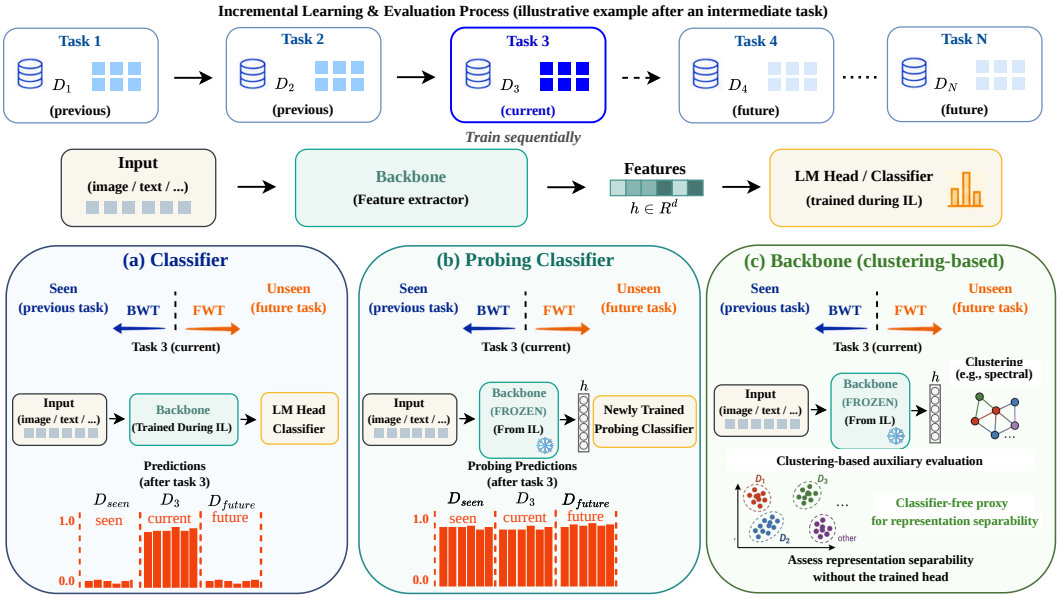


Figure 1: Overview of evaluating BWT and operational FWT at three interfaces in discriminative backbones. We measure transfer through the original classifier, a probing classifier, and a classifier-free proxy for representation separability. The backbone diagnostic does not replace task accuracy; it evaluates separability without using the sequentially trained prediction head. In generative backbones, label tokens replace [CLS] tokens and LM heads serve as the prediction interface. Details are in Appendix A.

If forgetting is mainly a head-level readout problem, then low classifier accuracy does not necessarily imply that the backbone has irreversibly lost the task. Conversely, if a newly trained probing classifier compensates for weak or shifted representations, it may understate representational forgetting.

Recent studies suggest that the backbone can retain more information than classifier-level accuracy indicates, and that much of the observed degradation can arise in the classifier or LM head [Davari et al., 2022, Zheng et al., 2024, 2025a]. Other work studies Forward Transfer (FWT) and argues that less forgetting can provide a useful inductive bias for future tasks [Chen et al., 2023, Zheng et al., 2025b]. These findings raise a more basic measurement question: when an incremental learner fails on old tasks, where is forgetting being measured?

We address this question by revisiting Backward Transfer (BWT) and FWT under three evaluation interfaces: the original classifier or LM head, a probing classifier, and the backbone representation. The main contribution is not a new metric that replaces accuracy. It is a diagnostic decomposition showing that standard IL metrics conflate three distinct estimands: deployed head behavior, recoverable information under a new readout, and classifier-free representation separability. Since the backbone has no standalone prediction interface, we approximate changes in backbone separability using clustering-based diagnostics.

Our analysis shows that the apparent relationship between BWT and FWT changes substantially across these interfaces. Classifier-level metrics often overstate forgetting because the head is itself unstable under sequential updates, while probing can understate forgetting because a newly trained classifier can compensate for shifted or weakened representations. Backbone diagnostics suggest intermediate representation drift and a consistent stability–plasticity pattern across clustering algorithms and metrics.

Motivated by this observation, we study Just LM-Head Tuning (JLT), a simple head-only adaptation procedure. JLT is not intended to be a new continual learning algorithm that solves the full IL setting. Instead, it is a diagnostic intervention: by re-optimizing only the LM head on top of a fixed incrementally trained backbone, we estimate how much performance can be recovered by correcting head–backbone mismatch. This reframing is important because head-only adaptation, especially

when performed before evaluating each task, should be interpreted as recoverable performance rather than direct standard CIL accuracy.

We evaluate this diagnostic view across five benchmark groups spanning seven datasets, encoder and decoder language backbones, multiple IL methods in CIL, and a discriminative-backbone TIL sanity check. Our main contribution is to formalize BWT and an operational FWT/plasticity-gain measure as interface-dependent quantities: the same formula answers different questions when measured through the trained head, a fresh probe, or a backbone-level diagnostic. Building on this formalization, we provide a three-interface diagnostic framework that separates deployed performance, readout recoverability, and classifier-free representation separability. Empirically, we show that these interfaces lead to qualitatively different conclusions, that the backbone-level stability–plasticity pattern is consistent across clustering algorithms and metrics, and that JLT can be used as a head-only intervention to estimate how much apparent forgetting is recoverable through readout re-alignment.

2 Metrics are interface-dependent

2.1 Problem setup

We consider an incremental learner that receives a sequence of tasks $D = \{D_1, D_2, \cdots, D_T\}$, where each task $D_t = \{(x_t^i, y_t^i)\}_{i=1}^{N_t}$ contains inputs $x_t^i \in X_t$ and labels $y_t^i \in Y_t$. The learner updates the same model sequentially from task 1 to task T.

The two most common IL scenarios are Class-Incremental Learning (CIL) and Task-Incremental Learning (TIL). In CIL, label sets from different tasks do not overlap, i.e., $Y_i \cap Y_j = \emptyset$ for $i \neq j$, and the model must predict among all classes observed so far. In TIL, the task identity is available at evaluation time, so prediction is restricted to the classes of the corresponding task. Because CIL requires a shared prediction space across tasks, it is usually more challenging than TIL [Tao et al., 2023]. We focus on CIL in the main results and use TIL as a sanity check in Section D.1.

2.2 Evaluation metrics for IL

BWT [Lopez-Paz and Ranzato, 2017, Ebrahimi et al., 2018, 2020] is a representative IL metric for *memory stability*. It compares performance on a previous task immediately after that task is learned with performance on the same task after the full task sequence has been learned:

$$BWT = \frac{1}{T-1} \sum_{i=1}^{T-1} (a_{T,i} - a_{i,i}) \tag{1}$$

where T is the last task, $a_{T,i}$ is the score on the i-th task after training up to the T-th task, and $a_{i,i}$ is the score on the i-th task immediately after training that task. The score a can be instantiated at different evaluation interfaces: the original classifier or LM head, a probing classifier, or a backbone-level representation diagnostic. These instantiations answer different questions and should not be treated as interchangeable.

FWT measures *learning plasticity*: how much the model improves on a task after actually training on it, relative to its state before seeing that task. This is harder to measure than BWT because a standard classifier often has no output units for unseen classes before the task is introduced. Prior work therefore often uses random accuracy as a proxy [Ke et al., 2020, Wołczyk et al., 2021, Ke and Liu, 2022, Wang et al., 2024] or trains a separate classifier to estimate performance on the next task [Chen et al., 2023]. In this work, we use the following operational form and instantiate it separately for each evaluation interface:

$$FWT = \frac{1}{T-1} \sum_{i=2}^{T} (a_{i,i} - a_{i-1,i}) \tag{2}$$

where T is the last task, $a_{i,i}$ is the score on the i-th task immediately after learning the i-th task, and $a_{i-1,i}$ is the score on the i-th task before training on it. For classifier-level evaluation, this captures the behavior of the prediction interface. For probe-level evaluation, it captures how much a newly trained classifier can recover from frozen representations. For representation-level evaluation, it approximates changes in backbone separability through classifier-free diagnostics.

Table 1: Evaluation interfaces considered in this paper. Each interface instantiates the same BWT/FWT formulas but answers a different measurement question.

Interface	What is frozen?	Readout	Measures	Does not measure
Sequential head	Nothing	Original IL classifier or LM head	Deployed IL performance through the trained prediction interface	Whether the backbone retained information
Probe	Backbone	Newly trained classifier	Recoverable information under a fresh supervised readout	Deployed performance of the IL model
Backbone clustering	Backbone	No supervised head	Classifier-free proxy for representation separability	Task accuracy or calibrated prediction
JLT	Backbone	LM head only	Recoverable performance under head re-alignment	Standard CIL performance

We use the term FWT in the operational sense common in continual learning evaluation, but the quantity in Eq. (2) should be read as a task-exposure plasticity gain. It measures how much the model improves on task i after training on task i, relative to the state immediately before seeing that task. This differs from purely zero-shot forward transfer. We keep the term FWT for consistency with IL literature, but throughout the paper we interpret it as an interface-specific plasticity-gain measure.

2.3 Three estimands and a head-only intervention

BWT and operational FWT are formulas over scores $a_{t,i}$, but the score itself is defined by an evaluation interface. Let $m_t = (f_t, g_t)$ denote the model state after training up to task t, where f_t is the backbone and g_t is the sequentially trained prediction head. An evaluation interface e specifies which components are frozen, which readout is used or fitted, what auxiliary data are allowed for fitting, and which scoring rule is applied. It induces an interface-specific score

$$a^e_{t,i} = S_e(m_t; D^{\mathrm{tr}}_i, D^{\mathrm{te}}_i),$$

where D^{tr}_i and D^{te}_i denote the task split available to the evaluation protocol. BWT and operational FWT are therefore not properties of m_t alone, but of the pair (m_t, e):

$$\mathrm{BWT}^e = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(a^e_{T,i} - a^e_{i,i} \right),$$

$$\mathrm{FWT}^e_{\mathrm{op}} = \frac{1}{T-1} \sum_{i=2}^{T} \left(a^e_{i,i} - a^e_{i-1,i} \right).$$

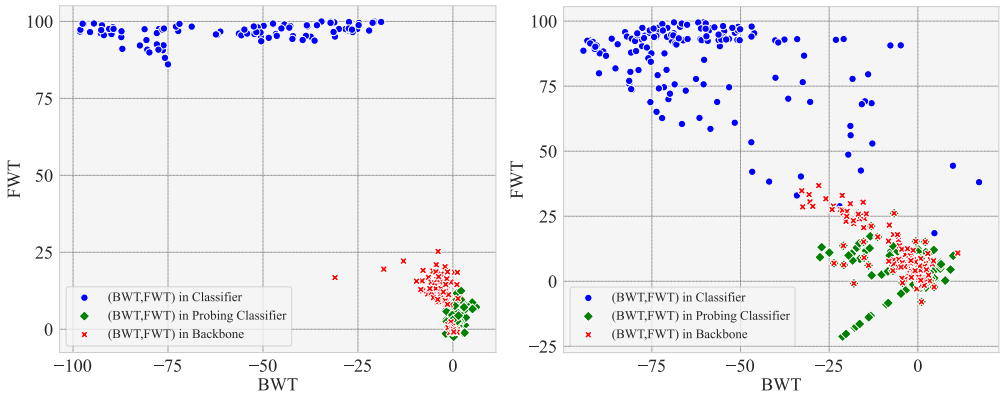
Reporting BWT or operational FWT without specifying e is therefore an underspecified measurement. Different interfaces can induce different signs and magnitudes because they instantiate different scoring functionals over the same model trajectory.

The sequential head measures deployed behavior, a probe measures recoverability under a newly optimized readout, and backbone clustering measures classifier-free separability. JLT adds a head-only counterfactual intervention for decoder models: it freezes the incrementally trained backbone and asks how much performance can be recovered by re-aligning the LM head.

3 Diagnostic interfaces

3.1 Classifier-level BWT and FWT

The most common evaluation interface is the classifier or LM head that is updated sequentially with the backbone. This is the correct interface for reporting task accuracy, but it can be difficult to interpret mechanistically. Prior work shows that sequentially trained classifiers can become strongly biased toward recently learned classes [Hou et al., 2019, Wu et al., 2019, 2022, Zheng et al., 2024]. As a result, low accuracy on earlier tasks may reflect instability in the prediction head, degradation in the backbone, or both.



(a) Discriminative backbone

(b) Generative backbone

Figure 2: BWT and operational FWT evaluated under three interfaces. Results are aggregated over the benchmark streams in Table 5; the number of incremental tasks varies by dataset.

Figure 2a illustrates this ambiguity. The classifier-level BWT values differ substantially across IL methods, suggesting different degrees of forgetting. However, because the classifier itself is part of the sequentially updated system, these values do not determine whether the retained representation has disappeared or whether it is no longer read out correctly. Classifier-level operational FWT is even less direct: before a new task is introduced, the classifier may not have a meaningful prediction interface for unseen classes. Thus, classifier-level BWT measures deployed performance but can conflate representation drift with head instability, while classifier-level FWT is difficult to interpret before unseen classes have a meaningful output interface.

3.2 Probe-level BWT and FWT

Another common interface trains a probing classifier on frozen backbone features, as shown in Figure 1. This removes the bias of the original sequential classifier and can indicate whether useful information remains accessible in the backbone [Hou et al., 2019, Zhou et al., 2022, Zheng et al., 2024]. However, probe-level evaluation answers a different question from standard task accuracy: it measures what a newly optimized classifier can recover from the representation, not what the incrementally trained model actually predicts.

In Figure 2a, probe-level BWT remains close to 0 for most models and IL methods. This suggests that much of the information needed for old tasks is still recoverable. At the same time, a strong probing classifier may compensate for representation shifts or weak separability [Zhou and Srikumar, 2021b], thereby masking changes in the backbone itself.

For FWT, the probing interface freezes the backbone after each task and trains a classifier over the relevant classes. This makes it possible to evaluate future-task labels, but the measured score depends heavily on the new classifier. Thus, probing is useful for measuring recoverability under a new readout, but it can mask representation shifts that remain recoverable to the probe.

3.3 Backbone-level diagnostic of BWT and FWT

The backbone has no standalone prediction interface, so backbone-level transfer cannot be observed through accuracy in the same way as classifier-level transfer. We therefore use clustering as an auxiliary, classifier-free diagnostic of representation separability. As shown in Figure 1, we extract backbone representations after each task, cluster them, align clusters to labels, and use the resulting scores to instantiate BWT-backbone and FWT-backbone. This diagnostic is not intended to replace task accuracy. Instead, it approximates how separable task representations remain independently of a learned prediction head.

This choice follows a broader line of work that analyzes backbone features to reduce dependence on classifier bias [Liu et al., 2019a, Zhou and Srikumar, 2021a,b]; related work also uses feature diversity as an auxiliary IL diagnostic [Chen et al., 2023]. We emphasize that clustering is not a

Table 2: Aggregate evidence for the BWT–FWT relation in the CIL scenario. The same algebraic BWT/FWT definitions are applied to all interfaces. Std summarizes variability across the listed diagnostic configurations and should not be interpreted as a standard error over independent samples. Full Pearson and Spearman tables are reported in Appendix E.

Interface	Diagnostic configurations	Mean Pearson r	Std	Range	Sign pattern
Trained head	10 backbones	-0.13	0.38	-0.52 to 0.43	Mixed
Probe	10 backbones	-0.19	0.44	-0.62 to 0.65	Mixed
Backbone diagnostic	150 non-independent diagnostic configurations	-0.74	0.13	-0.93 to -0.43	Negative in all diagnostic configurations

gold-standard measure of representation quality. It is sensitive to feature normalization, the number of clusters, class imbalance, cluster-label matching, the clustering algorithm, and the evaluation metric. This is why we do not use a single clustering result as a replacement for accuracy. Instead, we use clustering as a stress test: if the same qualitative trend appears across algorithms and metrics, then the trend is informative as a classifier-free diagnostic.

We evaluate clustering diagnostics with five algorithms and three metrics: ACC, ARI, and NMI. Main visualizations use Spectral Clustering, while Table 2 aggregates all clustering algorithms and metrics. Thus, backbone clustering is used only as a classifier-free separability proxy, not as a replacement for task accuracy. Details are in Appendix B.

4 Experimental setup

4.1 Task & baseline

We evaluate five benchmark groups spanning seven sentence-level datasets, as shown in Table 5. Each dataset defines its own incremental stream, with task counts ranging from 5 to 15. For text classification, we use three datasets, AGNews, DBPedia, and YaHoo, as Topic3Datasets [Zhang et al., 2015]. For intent classification, we use CLINC150 [Larson et al., 2019] and Banking77 [Casanueva et al., 2020]. Finally, for relation extraction, we use FewRel [Han et al., 2018] and TACRED [Zhang et al., 2017].

For discriminative backbones, we adopt four representative IL methods: Base, ER [Chaudhry et al., 2019], DER++ [Buzzega et al., 2020], and CLSER [Arani et al., 2022]. For generative backbones, we compare eight baselines: Base, DER++, CLSER, L2KD [Chuang et al., 2020], LAMOL_g, LAMOL_t [Sun et al., 2020], LAMOL_KD [Zheng et al., 2024], and PCLL [Zhao et al., 2022]. We also report recent competitive frozen-backbone methods, such as SEQ* [Zheng et al., 2024] and KLDA [Momeni et al., 2025], in the task-performance table. Because these methods keep the backbone fixed after warm-up or use fixed embeddings, their backbone-level BWT and FWT diagnostics are constant by construction, so they are not included in the backbone-diagnostic correlation analysis. For detailed descriptions of the baselines, please refer to Appendix C.

4.2 Backbone

We evaluate both discriminative backbones (encoder architectures) and generative backbones (decoder architectures). For discriminative backbones, we use BERT-base, BERT-large [Devlin et al., 2019], RoBERTa-base, and RoBERTa-large [Liu et al., 2019b]. For generative backbones, we use Pythia [Biderman et al., 2023] models based on GPT-NeoX [Black et al., 2022] in different sizes (70m, 160m, 410m), as well as Qwen2-0.5B [Yang et al., 2024a], Qwen2.5-0.5B [Qwen et al., 2025], and Qwen3-0.6B [Yang et al., 2025]. Due to resource limits, we cannot run every baseline on every larger model; for model-type and model-size analysis, we additionally evaluate the Base method on Pythia-1.4B/2.8B/6.9B, Llama3.2-1B/3B, Llama3.1-8B [Dubey et al., 2024], and Qwen3-0.6B/1.7B/4B/8B. More details on training and backbone evaluation are in Appendix A.1.

5 Results

5.1 The same BWT/FWT metric yields different conclusions

The three interfaces do not merely differ in magnitude; they support different qualitative conclusions. In discriminative backbones, the per-method plots in Appendix Figure 7 show that classifier-level

BWT suggests severe forgetting, especially for Base, because the sequential classifier is strongly affected by head bias. Probe-level BWT remains close to 0, suggesting that a newly trained classifier can recover much of the old-task information. Backbone-diagnostic BWT lies between these two views: the representation itself changes, but not as drastically as classifier-level evaluation implies.

The operational FWT results show the same interface dependence. Classifier-level FWT values can exceed 80, largely reflecting the introduction of task-specific output classes rather than pure transfer. Probe-level FWT remains close to 0 because the probe can compensate for representation differences. By contrast, FWT-backbone ranges over a meaningful intermediate scale, indicating measurable changes in representation separability before and after task training.

The generative-backbone results in Figure 2b and Appendix Figure 8 mirror this pattern. Classifier-level BWT again indicates severe forgetting, with many experiments scoring below -80, while the backbone diagnostic shows that decoder representations retain substantially more structure than the LM head can read out directly. This suggests that the backbone–head gap is not limited to encoder classifiers.

Finding 1. The trained head suggests catastrophic forgetting, the probe suggests strong recoverability, and the backbone diagnostic indicates partial representation drift. Forgetting is therefore not a single model-level number; it depends on the interface through which it is measured.

5.2 Backbone trends are consistent across clustering choices

A potential concern is that BWT and FWT share the intermediate term $a_{i,i}$, which can induce a negative association by construction:

$$BWT_i = a_{T,i} - a_{i,i}, \qquad FWT_i = a_{i,i} - a_{i-1,i}.$$

This algebraic coupling applies equally to all three interfaces. Therefore, the key question is not whether a negative correlation can occur, but whether the same trend appears uniformly across interfaces. It does not. As Table 2 shows, the trained head and probe interfaces have mixed signs and weak aggregate correlations, whereas the backbone diagnostic is negative in every clustering algorithm, metric, and backbone combination. The 150 diagnostic configurations are not statistically independent samples, since multiple clustering algorithms and metrics are applied to related representations. We use them to assess diagnostic robustness, not to estimate a sampling distribution. We therefore interpret the backbone trend as comparative, interface-specific diagnostic evidence, not as a consequence of the BWT/FWT formula alone.

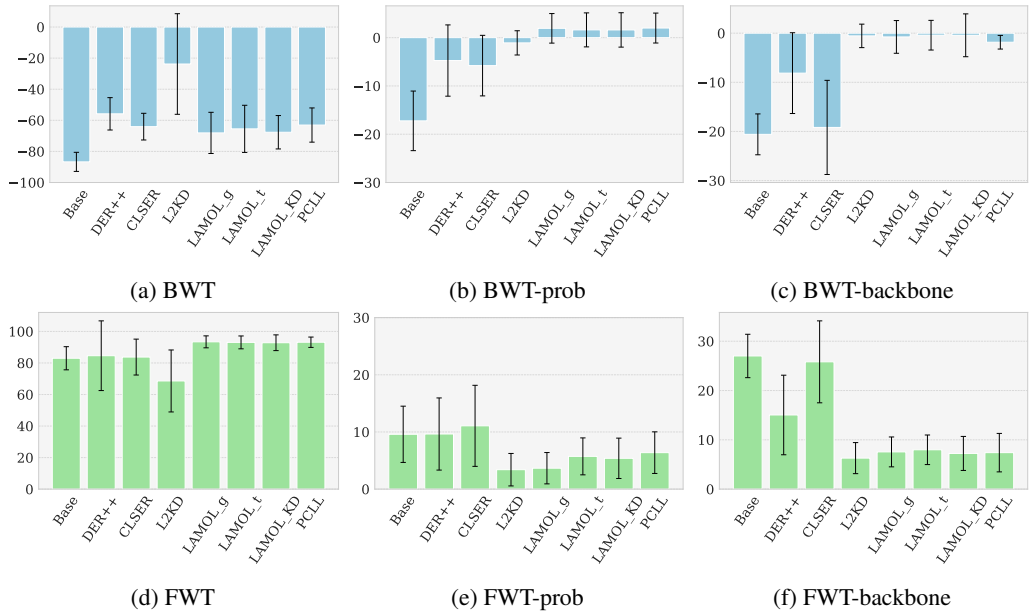
Consistency across five clustering algorithms and three metrics is important for our diagnostic interpretation: our conclusion does not depend on choosing Spectral Clustering or ACC. Full Pearson and Spearman correlations are reported in Tables 10 and 11. As a coupling-aware summary, we also consider $BWT + FWT = a_{T,i} - a_{i-1,i}$, which cancels the shared post-training term $a_{i,i}$. We do not use this as a primary metric, but it provides a useful check when interpreting BWT–FWT relations and the model-size analysis in Appendix E.

Finding 2. The representation-level stability–plasticity pattern is consistent across diagnostic choices, while the trained-head and probe interfaces remain unstable in sign and magnitude.

5.3 Stability-Plasticity Dilemma in Backbone

The Stability-Plasticity Dilemma was long regarded as a persistent challenge in neural networks, but recent IL methods have made it appear as though the problem has largely been resolved. However, this progress has mainly come from efforts to preserve the backbone as much as possible while adjusting the classifier or LM head. According to our observations, the dilemma still remains strong at the backbone level. As shown in Figure 8, there was a meaningful correlation: when BWT-backbone was high, FWT-backbone was also high, and when BWT-backbone was low, FWT-backbone was also low. This suggests that although recent IL methods have improved performance by reducing cost and preventing catastrophic forgetting through freezing the model, whether the backbone itself effectively learns knowledge still requires further reconsideration.

Finding 3. It is difficult to conclude that IL methods have fully overcome the Stability-Plasticity Dilemma at the backbone level.



(a) BWT (b) BWT-prob (c) BWT-backbone

(d) FWT (e) FWT-prob (f) FWT-backbone

Figure 3: Average and standard deviation of classifier-level, probe-level, and backbone-diagnostic BWT/FWT for each IL method in six generative backbones, aggregated over the benchmark streams in Table 5.

5.4 Secondary observation: scaling improves retention more than plasticity

The backbone diagnostic also provides a useful lens on scaling. In the Base setting, increasing model size within Pythia, LLaMA, and Qwen improves BWT-backbone more consistently than FWT-backbone, as shown in Figure 11. This suggests that scale helps retention more than it changes the immediate plasticity gain measured by our operational FWT. We treat this as a secondary observation, but it illustrates how the diagnostic can highlight trends that are difficult to see through the final head alone.

6 A Head-Only Intervention Estimates Recoverable Forgetting

Table 3: Aggregate JLT gap over three decoder backbones and five benchmark groups. The JLT numbers correspond to the task-adaptive diagnostic protocol, where the backbone is frozen and the LM head is re-aligned before evaluating each task. $G_{\mathrm{JLT}} = A_T^{\mathrm{JLT}} - A_T^{\mathrm{standard}}$ measures recoverable performance, not standard CIL performance.

IL method	**Avg standard** A_T	**Avg JLT** A_T	G_{JLT}	**Avg standard BWT**	**Avg JLT BWT**
Base	11.07	74.73	+63.65	-87.79	-19.12
L2KD	40.72	95.56	+54.84	-42.75	-1.76
LAMOL_KD	40.63	94.14	+53.50	-62.39	-2.68
PCLL	41.63	94.51	+52.88	-61.48	-1.89

Table 3 reports the aggregate JLT gap used in our head-only counterfactual intervention. The three-interface analysis suggests that poor standard accuracy may not imply that the backbone has erased the task. To test this interpretation, after an IL method has trained the model sequentially, we freeze the backbone and re-optimize only the LM head. This asks: how much of the lost task performance is recoverable through readout re-alignment alone?

The purpose of JLT is not to compete with standard IL protocols. Its purpose is to measure a counterfactual: if the incrementally trained backbone were fixed and only the readout were re-aligned,

how much performance would be recoverable? We therefore report the JLT gap,

$$G_{\mathrm{JLT}} = A_T^{\mathrm{JLT}} - A_T^{\mathrm{standard}},$$

as the primary quantity. A large gap indicates that standard accuracy loss is not fully explained by representation erasure; a small gap would indicate that the backbone itself has lost the relevant information.

In the diagnostic phase, we freeze the incrementally trained backbone and update only the LM head with a label-token cross-entropy objective. No adapters, external encoders, or independent classifiers are added. Implementation details are in Appendix F, and the full per-model/per-dataset table is reported in Appendix Table 12.

The aggregate results show that all four IL methods have large recoverable gaps. Although standard evaluation indicates severe forgetting, freezing the backbone and re-aligning only the LM head recovers much of the performance. For example, L2KD averages 40.72 under standard evaluation but 95.56 under the diagnostic head-realignment protocol, reducing average BWT from -42.75 to -1.76. Across the same full results, joint fine-tuning averages about 96.38 in A_T, while L2KD+JLT averages 95.56, motivating a diagnostic decomposition of the gap to the joint-training reference.

Let A_T^{joint} denote the joint-training reference. The gap from standard IL evaluation to the joint reference can be decomposed as

$$A_T^{\mathrm{joint}} - A_T^{\mathrm{standard}} = \underbrace{A_T^{\mathrm{JLT}} - A_T^{\mathrm{standard}}}_{\text{readout-recoverable gap}} + \underbrace{A_T^{\mathrm{joint}} - A_T^{\mathrm{JLT}}}_{\text{residual gap}}.$$

We use this decomposition only diagnostically. The first term estimates the portion of apparent forgetting that can be removed by re-aligning the LM head on a fixed incrementally trained backbone; the second term is the residual gap that head re-alignment alone does not remove. Thus, for methods such as L2KD, the task-adaptive JLT diagnostic leaves only a small residual gap relative to the joint-training reference, suggesting that a large part of the standard-evaluation failure is readout-recoverable under this oracle-style protocol.

These results should not be read as evidence that head-only tuning is a deployable replacement for continual learning. The task-adaptive JLT variant adapts the head before evaluating each task and therefore estimates recoverable performance rather than standard CIL accuracy. Its value is diagnostic: it suggests that many failures attributed to catastrophic forgetting are more precisely head–backbone mismatch plus partial representation drift.

Finding 4. JLT turns the diagnostic claim into an intervention. When the backbone is fixed and only the LM head is re-aligned, much of the apparent forgetting becomes recoverable.

Threats to interpretation. Our backbone diagnostic is not task accuracy; it is a classifier-free separability proxy. The clustering configurations are not independent statistical samples, but robustness stress tests over related diagnostic choices. Because BWT and operational FWT share $a_{i,i}$, we interpret their relation comparatively across interfaces rather than as standalone proof of a universal trade-off. Finally, task-adaptive JLT is oracle-style: it measures recoverability under head re-alignment, not deployable CIL performance.

7 Conclusion

Our central message is that forgetting in incremental learning is not a single model-level quantity. It is an interface-dependent measurement. The sequential head measures deployed behavior, the probe measures recoverability under a new readout, and the backbone diagnostic measures classifier-free separability. These views can disagree sharply, and treating them as interchangeable can lead to misleading conclusions about stability and plasticity. The JLT intervention shows why this distinction matters: when the incrementally trained backbone is fixed and only the LM head is re-aligned, much of the apparent forgetting is recoverable. This does not mean that standard IL is solved, nor that head-only tuning is a deployable replacement for continual learning. It means that many failures attributed to catastrophic forgetting are better understood as a combination of partial representation drift and readout mismatch. We therefore recommend that future incremental learning work report not only BWT, FWT, and average accuracy, but also the interface through which these quantities are measured.

Broader impacts

This work does not release a new pretrained model or dataset. Its positive impact is to make continual learning evaluation more transparent by separating deployed performance from recoverable representation information. By re-establishing BWT and FWT under classifier, probe, and backbone interfaces, we show that transfer metrics can behave fundamentally differently depending on where they are measured, enabling more principled and reproducible evaluation.

A possible negative impact is that diagnostic recoverability could be mistaken for deployable performance, leading to overoptimistic claims about model updating. We mitigate this by explicitly separating standard IL results from diagnostic JLT recoverability and by recommending that future work report the evaluation interface used for each metric.

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Limitations

We use clustering-based diagnostics to approximate changes in backbone separability, but this is not a complete substitute for classifier-level task accuracy. Clustering scores depend on the algorithm, number of clusters, feature normalization, class balance, cluster-label matching, and evaluation metric. Our goal is therefore not to propose a perfect new evaluation metric, but to provide an auxiliary view of how the backbone representation changes during IL.

JLT also has an important limitation: the task-adaptive variant performs head-only adaptation before evaluating each task. This should be interpreted as an oracle-style diagnostic of recoverable information in the backbone, not as a deployable CIL protocol that has access to each test task's training data at evaluation time. The reported gains are useful for quantifying head–backbone mismatch, but they should not be read as direct standard IL performance.

Incremental learning tasks that require sequential fine-tuning of models require substantial resources, which limits experiments with larger generative models. We tried to use Qwen3-32B [Yang et al., 2025], Llama3-70B [Dubey et al., 2024], and related models, but there were resource limitations and difficulties in applying each IL method equally. Models larger than 8B typically require PEFT [Houlsby et al., 2019, Hu et al., 2021, Han et al., 2024]; in that setting, several IL methods (L2KD, LAMOL, LAMOL_KD) did not work reliably. Our resources were limited to training models below 10B, and proposing a new IL method that optimizes the backbone-level stability–plasticity trade-off remains future work.

A Experimental details

A.1 Backbone details

For the discriminative backbone, learning and evaluation are performed using the [CLS] token features of the last hidden states [Ethayarajh, 2019]. At this time, the classifier is a linear layer that uses the output dimension of the backbone as the input dimension and the number of classes for each task as the output dimension. In the TIL task, the task is learned and evaluated from the classifier as is, and in the CIL task, the logits of the classifier for each task are concatenated and used. To use the generative backbone in sentence-level tasks, we use two types of prompts depending on the task type. For text and intent classification, we use the following prompt:

```
"Input sentence: {text}\n The Label: {label}{eos token}"
```

For relation extraction, we use the following prompt:

```
"Input sentence: {text}\n The relationship between {head entity} and {tail
entity} is {label}{eos token}"
```

With the above prompts, we use causal language modeling loss to optimize labeleos token [Zheng et al., 2024]. However, depending on the IL method, we additionally use Cross-Entropy loss, KL-Divergence loss, MSE loss, etc.

A.2 Implementation details

We use the following settings for the five benchmark groups. For Topic3Datasets, we applied a max len of 256 and 3 epochs for each incremental task, for FewRel and TACRED, we applied a max len of 128 and 5 epochs, and for CLINC150 and Banking77, we applied a max len of 64 and 5 epochs. We used a learning rate of 1×10^{-5} for each backbone and 1×10^{-3} for the classifier with the AdamW optimizer [Loshchilov and Hutter, 2017]. When fine-tuning a probing classifier with the frozen features of the backbone, we train for 20 epochs, and all classifiers use the logit of the linear layer.

We also used four NVIDIA RTX 3080 (VRAM 24G) and eight NVIDIA A5000 (VRAM 24G) for our experiments. All experiments were performed three times. Unless otherwise noted, reported means average over the three random seeds, and error bars in figures denote one standard deviation across seeds. Correlation coefficients are computed over the corresponding dataset–method units for each evaluation interface. The compute estimates in Table 4 are approximate ranges from completed runs

Table 4: Estimated compute resources for the reported experiments. Runtimes are approximate ranges from completed runs on our local GPUs and exclude failed preliminary experiments unless noted. The number of runs counts benchmark-stream/model/method/seed configurations used for the reported results.

Experiment group	GPU	Approx. runtime per run	# runs	Estimated GPU-hours
Discriminative CIL	RTX 3080 / A5000 24GB	0.5–3 h	≈ 24	≈ 12–72
Generative CIL, small models	A5000 24GB * 8	1–4 h	≈ 72	≈ 72–288
JLT diagnostic	A5000 24GB * 4	0.2–1 h	≈ 18	≈ 3.6–18
Model-size analysis	A5000 24GB * 8	2–8 h	≈ 24	≈ 48–192

Table 5: The statistics on selected sentence-level datasets for IL. **Tasks** is the number of incremental tasks for each dataset, **CIL Classes** is the number of test set classes in the CIL scenario, and **TIL Classes** is the number of test set classes evaluated for each task in the TIL scenario.

Task	Dataset	# Classes	# Tasks	# CIL Classes	# TIL Classes	# Training Instances	# Test Instances
Text Classification	Topic3Datasets	25	5	25	5	75000	46000
Intent Classification	CLINC150	150	15	150	10	15000	4500
	Banking77	77	7	77	11	7191	2800
Relation Extraction	FewRel	80	8	80	10	33600	11200
	TACRED	40	8	40	5	5909	1259

and are intended to support reproducibility planning rather than exact accounting. Other than that, we used the basic parameter settings recommended in each study for the incremental learning methods.

For assets whose public dataset cards do not specify a single explicit license, we use the datasets only under their commonly distributed research benchmark setting and cite the original sources. We do not redistribute the raw datasets.

A.3 Evaluation metrics

A.3.1 Average accuracy

The most basic method to evaluate Incremental Learning is to measure the performance of all tasks after learning the final task. Accordingly, Average Accuracy has been used as a representative metric in almost all studies. However, when learning and evaluating a total of N tasks, it is much more advantageous to maintain the performance of $N-1$ previous tasks than the performance of the current 1 task, so there is an aspect that IL methods focus excessively on retention. Average Accuracy in task t is defined as follows:

$$\mathcal{A}_t = \frac{1}{t}\sum_{i=1}^{t} a_{t,i} \tag{3}$$

where $a_{t,i}$ represents the accuracy of the model incrementally learned from task 1 to t on the test set of task i.

A.3.2 Average incremental accuracy

Average Incremental Accuracy calculates the average of Average Accuracy from 1 to the last T task. This metric can prevent performance from being evaluated only by A_T, but overall, it shows a similar trend to Average Accuracy. Since the previous task accuracy still has a high proportion, maintaining the accuracy of the previous task is good for the overall result.

$$\overline{\mathcal{A}} = \frac{1}{T}\sum_{t=1}^{T} A_t \tag{4}$$

B Clustering algorithm details

To evaluate the features extracted from the backbone, we employ five representative clustering algorithms: K-means [MacQueen, 1967], Gaussian Mixture Model [Dempster et al., 1977], Spectral Clustering [Ng et al., 2001], Agglomerative Clustering [Johnson, 1967], and Deep Clustering [Xie

Table 6: Existing datasets and pretrained model assets used in this work. We cite the original papers in the main text and summarize the public license or access terms reported by the corresponding source or model card.

Asset	Type	Version/source	License or access terms
Topic3Datasets: AGNews, DBPedia, Yahoo	Dataset	Zhang et al. topic-classification datasets; DBpedia source data from DBpedia releases	AGNews/Yahoo: research-use terms from original sources; public dataset cards do not consistently specify a license. DBpedia: CC-BY-SA/GFDL terms for DBpedia releases.
CLINC150	Dataset	UCI Machine Learning Repository, `https://archive.ics.uci.edu/dataset/570/clinc150`	Creative Commons Attribution 4.0 International (CC BY 4.0).
Banking77	Dataset	PolyAI-LDN task-specific datasets / Hugging Face dataset card, `https://huggingface.co/datasets/PolyAI/banking77`	Creative Commons Attribution 4.0 International (CC BY 4.0).
FewRel	Dataset	THUNLP FewRel release / Hugging Face dataset card, `https://huggingface.co/datasets/thunlp/few_rel`	MIT License.
TACRED	Dataset	Stanford TACRED / LDC2018T24, `https://catalog.ldc.upenn.edu/LDC2018T24`	LDC User Agreement; access through Linguistic Data Consortium terms.
BERT	Model	Google BERT model card, e.g., `https://huggingface.co/google-bert/bert-base-uncased`	Apache License 2.0.
RoBERTa	Model	FacebookAI RoBERTa model card, e.g., `https://huggingface.co/FacebookAI/roberta-base`	MIT License.
Pythia / GPT-NeoX	Model	EleutherAI Pythia model cards, e.g., `https://huggingface.co/EleutherAI/pythia-410m-v0`	Apache License 2.0.
Qwen2, Qwen2.5, Qwen3	Model	Qwen model cards, e.g., `https://huggingface.co/Qwen/Qwen3-0.6B`	Apache License 2.0.
LLaMA 3.1/3.2	Model	Meta LLaMA model cards, e.g., `https://huggingface.co/meta-llama/Llama-3.2-3B`	Meta LLaMA Community License and Acceptable Use Policy.

et al., 2016]. While all algorithms are evaluated, the main visualizations use Spectral Clustering and the aggregate summary in Table 2 uses all algorithms and metrics.

B.1 Clustering evaluation metrics

ACC (Accuracy)

A contingency matrix is created to map the results from each clustering algorithm to the actual labels. The Hungarian algorithm is used on the contingency matrix to find the optimal mapping with the actual labels and evaluate the performance Hubert and Arabie [1985].

ARI (Adjusted Rand Index)

ARI measures the agreement between the clustering results and the actual labels, and compensates for the random prediction results. ARI is calculated based on pairs of data points. It compares the correspondence between two clusterings for each pair of data points. It has a scale between -1 and 1, where 1 indicates perfect agreement, 0 indicates agreement at the level of random clustering, and -1 indicates agreement lower than expected (random clustering) [Vinh et al., 2009, Zhang et al., 2019].

$$ARI = \frac{2 \cdot (TP \cdot TN - FN \cdot FP)}{(TP + FN) \cdot (FN + TN) + (TP + FP) \cdot (FP + TN)} \tag{5}$$

TP(True Positive): The number of sample pairs that belong to the same cluster in both clusterings.

TN(True Negative): The number of sample pairs that belong to different clusters in both clusterings.

FP(False Positive): The number of sample pairs that belong to the same cluster in one clustering, but to different clusters in the other clustering.

FN(False Negative): The number of sample pairs that belong to different clusters in one clustering, but to the same cluster in the other clustering.

NMI (Normalized Mutual Information)

Table 7: Results of the ER method in the RoBERTa-base model. A_T is the average accuracy by the classifier for the test of all tasks after incremental learning up to the last T-th task.

Scenario:CIL		Task				
Clustering Algorithm	Metric	Tacred	Banking77	Clinc150	Fewrel	Topic3
K-means	Acc	52.04 ± 1.5	65.23 ± 2.0	78.37 ± 1.8	62.29 ± 2.2	66.50 ± 1.7
	ARI	54.90 ± 1.3	71.68 ± 1.7	83.91 ± 2.0	66.18 ± 1.5	74.55 ± 1.8
	NMI	71.71 ± 1.4	80.55 ± 1.9	87.32 ± 2.1	77.91 ± 2.0	84.35 ± 1.6
GMM	Acc	52.06 ± 1.6	65.77 ± 1.8	78.90 ± 2.0	61.44 ± 2.1	61.70 ± 1.7
	ARI	57.62 ± 1.5	74.76 ± 1.9	81.73 ± 2.1	64.29 ± 1.8	73.53 ± 1.6
	NMI	71.15 ± 1.3	83.75 ± 2.0	88.40 ± 2.2	76.90 ± 1.9	82.16 ± 1.5
Spectral	Acc	59.82 ± 1.4	67.92 ± 2.0	80.02 ± 1.7	60.18 ± 1.9	61.80 ± 1.8
	ARI	55.40 ± 1.5	72.18 ± 1.8	83.18 ± 2.1	60.40 ± 2.0	76.23 ± 1.6
	NMI	70.17 ± 1.6	82.95 ± 1.9	89.47 ± 2.2	75.73 ± 2.0	84.68 ± 1.7
Agglomerative	Acc	63.82 ± 1.7	70.81 ± 1.8	79.69 ± 2.1	63.94 ± 1.9	67.28 ± 1.6
	ARI	61.30 ± 1.4	75.57 ± 2.0	84.34 ± 2.2	67.95 ± 1.8	77.96 ± 1.7
	NMI	72.20 ± 1.5	85.12 ± 2.1	90.57 ± 2.3	78.55 ± 2.0	84.56 ± 1.8
Deep_Clustering	Acc	59.58 ± 1.6	67.36 ± 1.9	80.45 ± 2.1	63.18 ± 1.7	61.44 ± 1.8
	ARI	61.62 ± 1.5	74.27 ± 1.8	84.02 ± 2.0	66.50 ± 1.6	73.54 ± 1.9
	NMI	70.51 ± 1.4	83.44 ± 2.0	88.79 ± 2.3	78.11 ± 2.1	83.66 ± 1.7
Classifier	A_T	49.19 ± 1.3	52.79 ± 1.8	66.69 ± 2.0	42.54 ± 1.9	73.23 ± 1.6

NMI evaluates the mutual information between the mapped cluster labels and actual labels, normalized to account for class imbalances and size differences. Since NMI is relatively insensitive to imbalances, it is the most consistent metric for evaluating high-dimensional models and multi-class scenarios. NMI range from 0 to 1, with values closer to 1 indicating high mutual dependency between labels.

$$I(U,V) = \sum_{u \in U} \sum_{v \in V} P(u,v) \cdot \log \left(\frac{P(u,v)}{P(u) \cdot P(v)} \right) \tag{6}$$

$$H(U) = - \sum_{u \in U} P(u) \cdot \log P(u) \tag{7}$$

$$NMI(U,V) = \frac{I(U,V)}{\sqrt{H(U)H(V)}} \tag{8}$$

$I(U,V)$ is the mutual information between the cluster set U and the true label set V. $H(U)$ is the entropy for each cluster and label. $NMI(U,V)$ is the mutual information $I(U,V)$ normalized by the geometric mean of the entropies of U and V.

B.2 Experiment results by clustering algorithm

We evaluated all models and IL methods with five clustering algorithms and three metrics. Most clustering algorithms recorded similar performance. For comparison with the commonly measured Average Accuracy by the classifier, we present the results of experiments with ER method on RoBERTa-base model as a representative of discriminative backbone in Table 7. Similarly, we present the results of experiments with LAMOL_g method on Pythia-410m model as a representative of generative backbone in Table 8. A_T is the performance on the last task T in the Average Accuracy presented in Appendix A.3.1.

In the CIL scenario, the clustering diagnostics often produce higher separability scores than the sequential classifier's A_T in terms of NMI, ARI, and Acc. In particular, clustering Acc is higher than A_T for four datasets (Tacred, Banking77, Clinc150, and Fewrel) with more than seven incremental tasks. On the other hand, in Topic3Dataset with only five incremental tasks, A_T is higher than the Acc of all clustering algorithms. This suggests that as the number of incremental tasks increases,

Table 8: Results of the LAMOL_g method in the Pythia-410m model. A_T is the average accuracy by the classifier for the test of all tasks after incremental learning up to the last T-th task.

Scenario:CIL		Task				
Clustering Algorithm	Metric	Tacred	Banking77	Clinc150	Fewrel	Topic3
K-means	Acc	27.36 ± 3.2	67.13 ± 2.8	71.39 ± 4.5	57.23 ± 3.1	63.23 ± 2.4
	ARI	24.66 ± 2.6	71.23 ± 3.1	76.24 ± 4.2	62.12 ± 2.9	68.22 ± 2.7
	NMI	45.67 ± 2.3	81.93 ± 3.0	85.25 ± 3.8	74.33 ± 4.5	80.91 ± 3.1
GMM	Acc	27.77 ± 2.8	65.38 ± 3.4	71.89 ± 3.7	58.36 ± 4.2	59.27 ± 2.5
	ARI	23.97 ± 3.1	70.46 ± 2.6	77.82 ± 4.1	61.99 ± 3.7	72.23 ± 3.0
	NMI	46.03 ± 3.4	82.23 ± 2.9	84.91 ± 3.3	73.10 ± 4.4	81.85 ± 3.5
Spectral	Acc	28.74 ± 4.1	66.40 ± 2.5	71.44 ± 3.6	55.13 ± 3.9	48.80 ± 2.8
	ARI	21.46 ± 3.7	69.45 ± 2.7	76.61 ± 4.4	56.08 ± 4.3	73.90 ± 3.1
	NMI	46.38 ± 3.2	82.42 ± 2.8	86.06 ± 4.1	72.96 ± 3.5	84.48 ± 2.7
Agglomerative	Acc	28.30 ± 3.1	68.12 ± 2.6	72.16 ± 3.8	61.79 ± 3.7	76.09 ± 3.9
	ARI	24.00 ± 2.7	71.44 ± 3.5	78.11 ± 3.3	65.40 ± 3.1	80.59 ± 4.2
	NMI	46.99 ± 2.9	83.86 ± 2.4	87.65 ± 4.0	76.31 ± 4.1	83.22 ± 3.5
Deep_Clustering	Acc	28.12 ± 2.8	66.32 ± 3.6	71.23 ± 4.2	56.12 ± 3.5	65.42 ± 4.3
	ARI	24.34 ± 3.3	69.23 ± 3.4	77.24 ± 3.7	63.78 ± 4.1	71.10 ± 3.6
	NMI	46.23 ± 2.6	83.23 ± 3.5	86.11 ± 3.9	74.60 ± 4.2	82.12 ± 3.3
Classifier	A_T	28.95 ± 3.2	51.95 ± 4.3	34.38 ± 3.1	23.09 ± 4.2	74.65 ± 3.4

classifier-level average accuracy can become increasingly affected by head-level forgetting. The clustering results should not be interpreted as higher task accuracy, but as evidence that representation separability can remain higher than what the sequential classifier exposes.

Our goal in this study is not to determine which clustering algorithm is best or which metric is the best. Therefore, we used the Spectral Algorithm, which showed average performance, for visualization and analysis in the main paper, and all the measurement results are presented in Appendix E.

C Incremental learning methods

We measured BWT and FWT for representative Incremental Learning methods. Aside from a brief explanation, we adopted detailed experimental settings widely used in prior studies.

Base - Sequentially fine-tunes tasks. Typically, when evaluating performance through a classifier, it is known to predict only the classes of the most recently trained task for all tasks.

ER[Chaudhry et al., 2019] - A classical replay technique that incorporates old samples to improve retention. When learning a new task, a portion of old samples is included in the training process. In this study, we include one old sample per class during training.

DER++[Buzzega et al., 2020] - DER++ extends ER by utilizing Knowledge Distillation through the MSE loss between a teacher model and a student model, rather than simply including old samples in training. Although originally developed for the computer vision domain, we evaluated this method with both discriminative and generative backbones.

CLSER[Arani et al., 2022]. CLS-ER builds on ER and DER++ by employing a dual-memory experience replay mechanism with fast and slow models. Like DER++, this method was initially used in the computer vision domain, but we applied it to both discriminative and generative backbones.

L2KD[Chuang et al., 2020] - L2KD is a method based on LAMOL, incorporating Knowledge Distillation. The teacher model learns the respective tasks first.

LAMOL_g & LAMOL_t[Sun et al., 2020] LAMOL is a method designed for generative models. When training on a new task, the model generates pseudo-samples of previous tasks and learns them together with the new task. The key difference is that LAMOL_g does not use the gen_token, whereas LAMOL_t does.

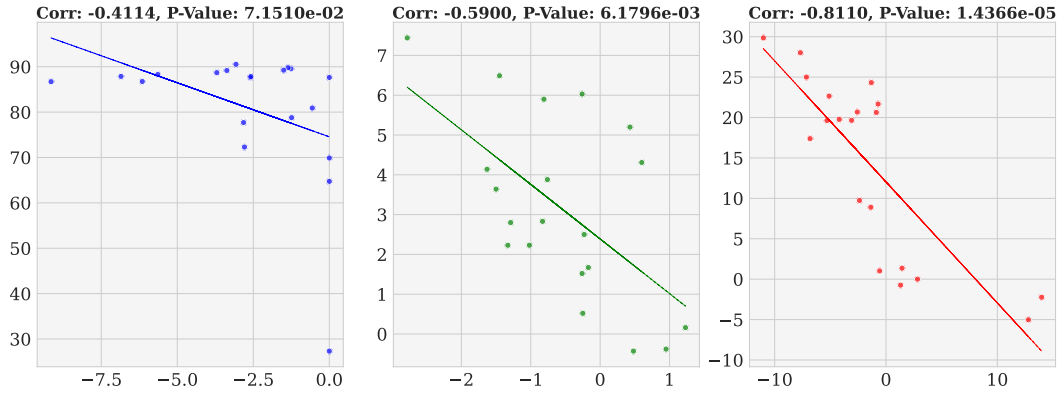


Figure 4: Pearson correlation coefficients of three BWT, FWT measurement methods of the RoBERTa-base model in the TIL scenario. Four IL methods were applied: Base, ER, DER++, and CLSER.

LAMOL_KD[Zheng et al., 2024] - Similar to L2KD, but with the distinction that the teacher model learns all previous tasks and transfers the knowledge via Knowledge Distillation.

PCLL[Zhao et al., 2022] - Based on LAMOL, PCLL introduces the use of Variational AutoEncoders (VAEs) to perform Knowledge Distillation.

SEQ*[Zheng et al., 2024] - SEQ* reduces backbone forgetting and classifier bias by freezing the backbone after warming up on the first task and training only the classifier for subsequent tasks using frozen backbone outputs. Since the backbone remains fixed after the first task, its representation-level diagnostic scores do not change. This makes SEQ* useful as a frozen-backbone baseline, but its BWT-backbone and FWT-backbone are always 0 by construction.

KLDA[Momeni et al., 2025] - KLDA projects input data into a high-dimensional feature space using a kernel function and then performs Linear Discriminant Analysis (LDA) to maximize class separability. This approach effectively captures non-linear decision boundaries and is used in continual learning to enhance class distinction based on embeddings extracted from foundation models.

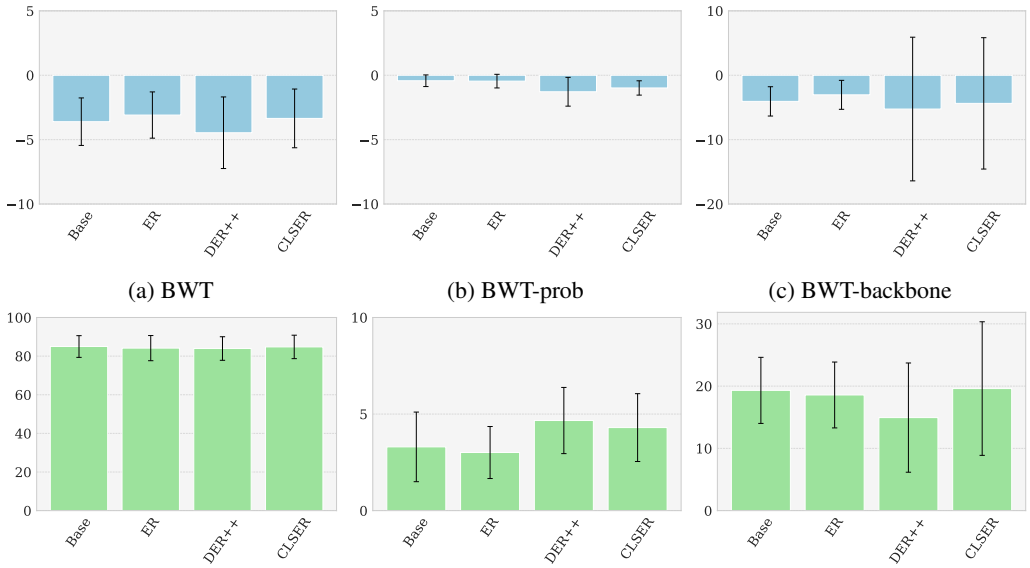
D Results in the TIL scenario

We compare the results of three measurement methods for BWT and FWT using four Incremental Learning methods across four discriminative backbones, as in the CIL scenario. As shown in Figure 4, the BWT-backbone and FWT-backbone measurement consistently recorded the highest negative correlation across all four backbones. Even in the TIL scenario, where separate classifiers are used for each task, the stability-plasticity dilemma appeared more prominently in the backbone than in the linear layer classifier.

Observing the x-axis in Figure 6, BWT values are close to 0 in the TIL scenario, as separate classifiers are used for each task. However, prior research [Zhou and Srikumar, 2021b] has suggested that classifiers can account for much of the performance, even when the model's representation is relatively weak. In comparison, BWT-backbone demonstrates a much broader range of values, providing evidence of differences in model performance and the varying effectiveness of incremental learning methods that are less visible through classifier performance. BWT-prob, as in the CIL scenario, measures values close to 0 for all models and methods due to the use of separate classifiers.

On the y-axis of Figure 6, the operational FWT results for the three measurement methods show distinct differences even in the TIL scenario. Classifier-level FWT values exceed 60 in most cases, as they are measured on classifiers that do not know the next class, similar to the CIL scenario. Lastly, FWT-backbone suggests performance improvements in the backbone as new tasks are learned, which aligns with the explanation provided for the CIL scenario in Section 3.3.

In all cases of Figures 5a, 5b, and 5c, the BWT measurement results follow the same order: ER > BASE > CLSER > DER++. (Of course, BWT-prob is trained separately on the classifier and is not learned together with the backbone.) In the TIL scenario, task-specific classifiers reduce



(a) BWT (b) BWT-prob (c) BWT-backbone

(d) FWT (e) FWT-prob (f) FWT-backbone

Figure 5: BWT, FWT measurement results by IL method of discriminative backbone in TIL scenario. Each figure shows the mean and standard deviation.

global class-incremental classifier bias, but the classifier and probe interfaces still depend on their readout mechanisms. BWT-backbone produced results very similar to the BWT measurements in the TIL scenario, suggesting that separability-based forgetting can be approximated without using the sequentially trained classifier.

Figures 5d, 5e, and 5f compare the operational FWT measurement results in the TIL scenario. In Figure 5d, classifier-level FWT scores all averaged above 80, showing little difference across methods. In Figure 5e, FWT-prob recorded extremely low values, averaging below 5 for all methods, with no discernible distinction. In Figure 5f, similar to Figure 7f, the results resemble those of the CIL scenario, as the backbone undergoes the same processes except for the difference in the classifiers used for CIL and TIL. This supports our interpretation that FWT-backbone measures backbone plasticity more independently of the classifier interface. All experimental results are presented in Table 9.

In the TIL scenario on the generative backbone, it was difficult to conduct experiments under the same conditions because some methods were optimized only for CIL. Many IL methods focus on CIL scenarios, which are much more challenging than TIL, where Base methods already perform well, using separate classifiers for each task.

D.1 TIL sanity check: not only CIL classifier bias

A natural concern is that the observed discrepancy is only a byproduct of class-incremental classifier bias. We therefore repeat the analysis in the TIL setting, where task identity restricts the prediction space. Figure 6 shows that the interface discrepancy remains: classifier and probe evaluations become less dominated by global class competition, but the backbone diagnostic still shows the most consistent stability–plasticity pattern among the evaluated interfaces. This supports our interpretation that the backbone–head gap is broader than class-incremental classifier bias.

E Full results

E.1 Per-method CIL interface plots

Figures 7 and 8 report the per-method BWT/FWT bar plots corresponding to the main text discussion. They show the same qualitative pattern summarized in Figure 2 and Table 2: the trained head suggests

Table 9: Pearson correlation coefficient between BWT and operational FWT for each metric in the TIL scenario. The results aggregate the benchmark streams in Table 5 for four methods: Base, ER, DER++, and CLSER.

Scenario:TIL		Discriminative backbone			
	Metric	BERT-base	BERT-large	RoBERTa-base	RoBERTa-large
Classifier	Acc	-0.4578	-0.6123	-0.5087	-0.6342
Prob Classifier	Acc	-0.5471	-0.4982	-0.4723	-0.6234
K-means	Acc	-0.8421	-0.7967	-0.8732	-0.8054
	ARI	-0.8845	-0.8123	-0.7987	-0.8698
	NMI	-0.8834	-0.8291	-0.8415	-0.8226
GMM	Acc	-0.8023	-0.8812	-0.8654	-0.7841
	ARI	-0.8912	-0.8764	-0.7998	-0.8457
	NMI	-0.8781	-0.8534	-0.8176	-0.8394
Spectral	Acc	-0.8107	-0.8723	-0.8321	-0.7955
	ARI	-0.8699	-0.8456	-0.8931	-0.8744
	NMI	-0.8543	-0.8198	-0.8411	-0.8294
Agglomerative	Acc	-0.8742	-0.8543	-0.7987	-0.8921
	ARI	-0.9112	-0.8763	-0.8194	-0.8774
	NMI	-0.8915	-0.8381	-0.8621	-0.8044
Deep Clustering	Acc	-0.8142	-0.7921	-0.8764	-0.8321
	ARI	-0.8776	-0.8221	-0.8442	-0.8167
	NMI	-0.8794	-0.8472	-0.8123	-0.8917

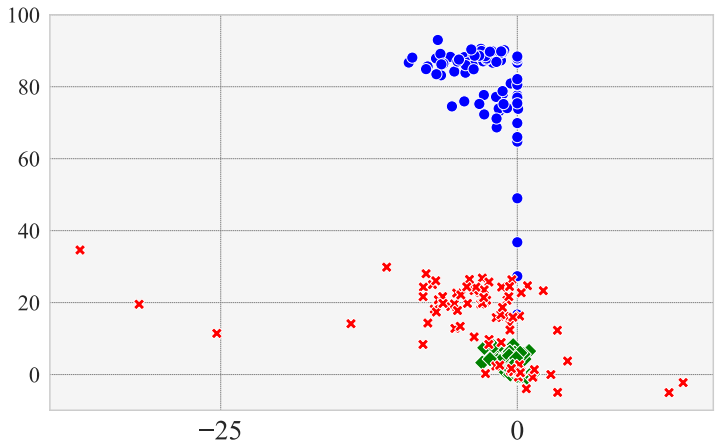
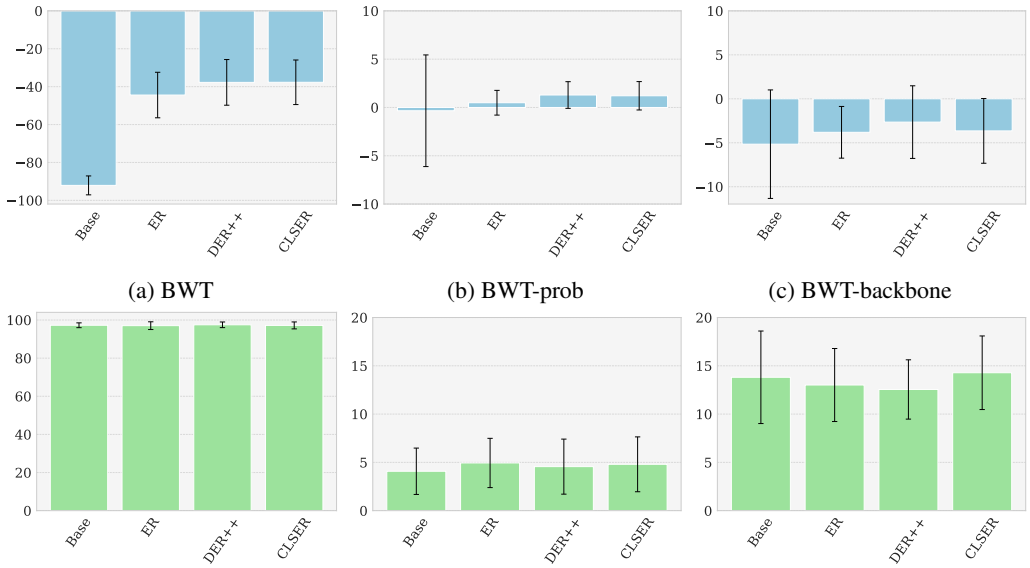


Figure 6: TIL sanity check. The x-axis represents BWT and the y-axis represents operational FWT; colors denote the three evaluation interfaces. Even when task identity restricts the prediction space, the interfaces remain separated, supporting that the backbone–head gap is not only CIL classifier bias.

severe forgetting, the probe suggests high recoverability, and the backbone diagnostic lies between these two views.

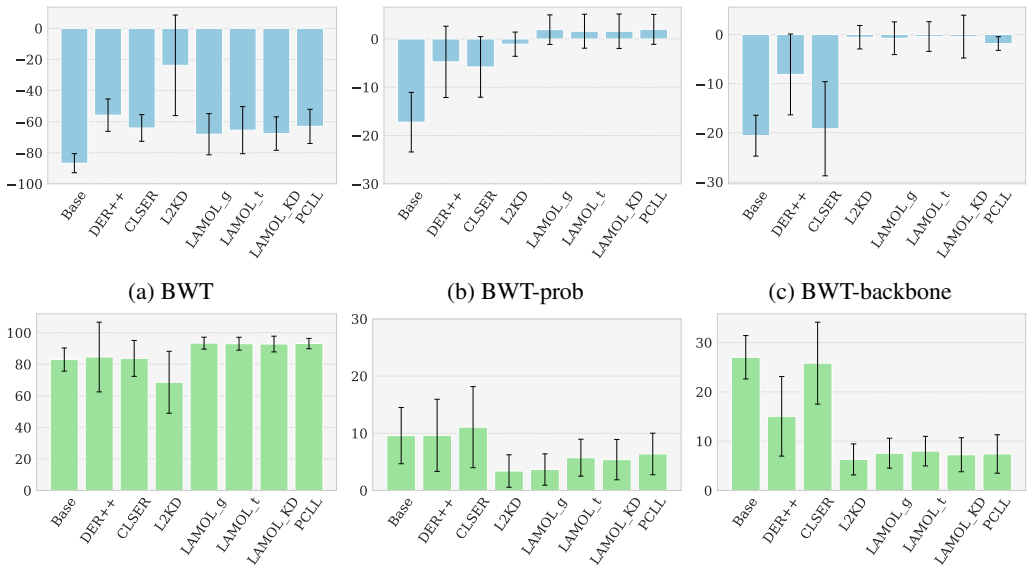
We conducted the same experiment for all clustering algorithms and metrics other than those analyzed in the main paper. Looking at the results measured in the discriminative backbone in Tables 10 and 11, the BWT and operational FWT correlations measured through the classifier and probing classifier have mixed signs and weaker aggregate magnitude than the backbone diagnostic. This shows that the stability–plasticity conclusion is interface-dependent rather than a direct consequence of the metric formula.



(a) BWT (b) BWT-prob (c) BWT-backbone

(d) FWT (e) FWT-prob (f) FWT-backbone

Figure 7: Average and standard deviation of classifier-level, probe-level, and backbone-diagnostic BWT/FWT for each IL method in four discriminative backbones, aggregated over the benchmark streams in Table 5.



(a) BWT (b) BWT-prob (c) BWT-backbone

(d) FWT (e) FWT-prob (f) FWT-backbone

Figure 8: Average and standard deviation of classifier-level, probe-level, and backbone-diagnostic BWT/FWT for each IL method in six generative backbones, aggregated over the benchmark streams in Table 5.

On the other hand, the BWT-backbone and FWT-backbone results show a negative correlation regardless of whether Acc, ARI, or NMI is used. The generative-backbone results in Tables 10 and 11 show the same trend. The trained-head and probing interfaces show weaker or mixed correlations, while all clustering algorithms and metrics show a consistently negative relation. Because the same shared-term structure applies to every interface, we use this pattern as comparative diagnostic evidence rather than as standalone proof of a universal trade-off.

Table 10: Pearson correlation coefficient between BWT and operational FWT for each metric in the CIL scenario.

Scenario:CIL		Discriminative backbone				Generative backbone					
	Metric	BERT-b	BERT-l	RoBERTa-b	RoBERTa-l	Pythia-70m	Pythia-160m	Pythia-410m	Qwen2-0.5B	Qwen2.5-0.5B	Qwen3-0.6B
Classifier	Acc	0.4272	0.1398	0.3807	0.2429	-0.5039	-0.3631	-0.3738	-0.5201	-0.3624	-0.3790
Prob Classifier	Acc	0.6478	0.1130	0.2468	0.1079	-0.4180	-0.4845	-0.6243	-0.4312	-0.4823	-0.6187
K-means	Acc	-0.6281	-0.5621	-0.4811	-0.5233	-0.8463	-0.8721	-0.8513	-0.8520	-0.8698	-0.8576
	ARI	-0.6638	-0.5132	-0.5314	-0.5419	-0.7992	-0.8103	-0.8192	-0.8021	-0.8135	-0.8250
	NMI	-0.6798	-0.6120	-0.5822	-0.6788	-0.8193	-0.9147	-0.8921	-0.8222	-0.9080	-0.8953
GMM	Acc	-0.5183	-0.5712	-0.4956	-0.4278	-0.7689	-0.7493	-0.7742	-0.7705	-0.7512	-0.7790
	ARI	-0.6762	-0.6888	-0.6232	-0.5145	-0.8101	-0.8018	-0.8439	-0.8124	-0.8040	-0.8471
	NMI	-0.6886	-0.6982	-0.6123	-0.6121	-0.7914	-0.8327	-0.8431	-0.7933	-0.8354	-0.8460
Spectral	Acc	-0.6918	-0.6128	-0.6157	-0.5987	-0.8102	-0.8333	-0.8688	-0.8118	-0.8305	-0.8650
	ARI	-0.6841	-0.6233	-0.6822	-0.6522	-0.8239	-0.8484	-0.8129	-0.8260	-0.8452	-0.8154
	NMI	-0.7213	-0.6434	-0.6557	-0.6857	-0.8231	-0.8923	-0.8725	-0.8257	-0.8894	-0.8699
Agglomerative	Acc	-0.4744	-0.5114	-0.6144	-0.4566	-0.7718	-0.7466	-0.6999	-0.7740	-0.7481	-0.7020
	ARI	-0.7413	-0.6912	-0.5989	-0.7362	-0.8654	-0.8132	-0.8948	-0.8623	-0.8180	-0.8905
	NMI	-0.9091	-0.7122	-0.7321	-0.7487	-0.8835	-0.9263	-0.8726	-0.8810	-0.9230	-0.8752
Deep Clustering	Acc	-0.5002	-0.4872	-0.6166	-0.5871	-0.7943	-0.8221	-0.7849	-0.7971	-0.8195	-0.8802
	ARI	-0.4748	-0.5237	-0.5891	-0.4824	-0.8412	-0.8109	-0.8019	-0.8390	-0.8137	-0.8065
	NMI	-0.6258	-0.6413	-0.6709	-0.6235	-0.8741	-0.8323	-0.8824	-0.8720	-0.8350	-0.8922

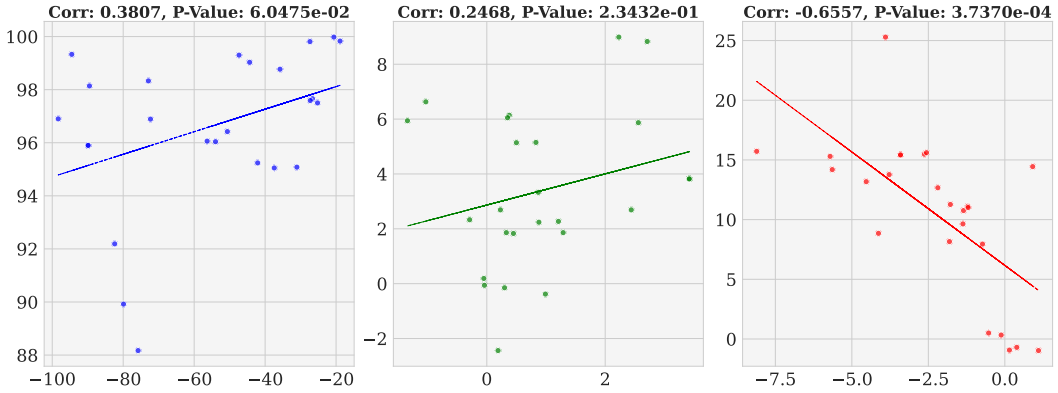


Figure 9: Pearson correlation coefficients between BWT and FWT by three evaluation methods in the RoBERTa-base model.

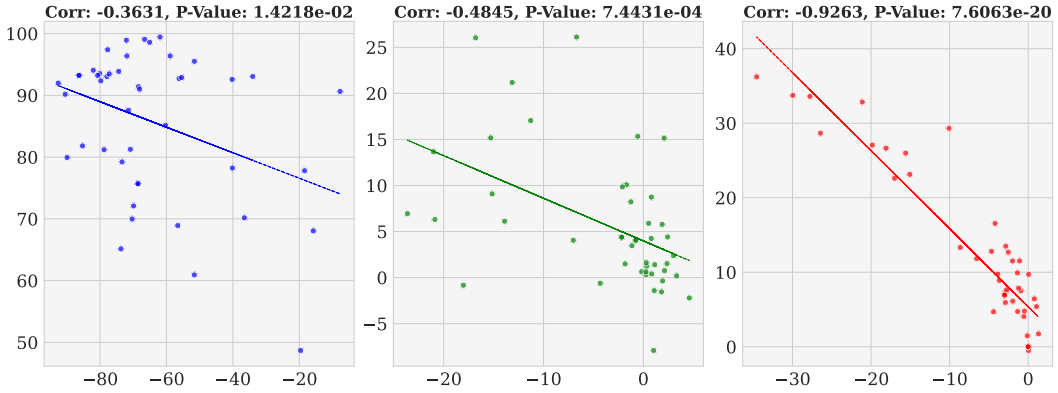


Figure 10: Pearson correlation coefficients between BWT and FWT by three evaluation methods in the Pythia-160m model.

E.2 Full JLT results

Table 12 reports the full per-model and per-dataset JLT results corresponding to the aggregate gap summarized in Table 3.

Table 11: Spearman correlation coefficient between BWT and operational FWT for each metric in the CIL scenario.

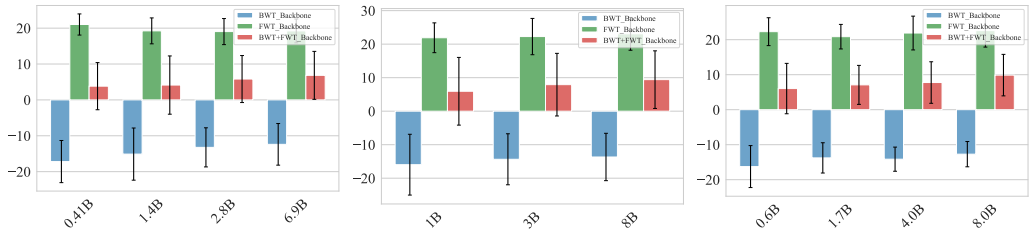
Scenario:CIL		Discriminative backbone				Generative backbone					
	Metric	BERT-b	BERT-l	RoBERTa-b	RoBERTa-l	Pythia-70m	Pythia-160m	Pythia-410m	Qwen2-0.5B	Qwen2.5-0.5B	Qwen3-0.6B
Classifier	Acc	0.4225	0.1327	0.3752	0.2385	-0.5142	-0.3729	-0.3654	-0.5300	-0.3800	-0.3670
Prob Classifier	Acc	0.6402	0.1063	0.2417	0.1046	-0.4239	-0.4725	-0.6339	-0.4350	-0.4870	-0.6280
K-means	Acc	-0.6246	-0.5587	-0.4873	-0.5182	-0.8368	-0.8660	-0.8481	-0.8420	-0.8580	-0.8550
	ARI	-0.6612	-0.5196	-0.5275	-0.5441	-0.8125	-0.8001	-0.8257	-0.8150	-0.8080	-0.8280
	NMI	-0.6459	-0.6073	-0.5792	-0.6741	-0.8204	-0.9212	-0.8874	-0.8250	-0.9120	-0.8930
GMM	Acc	-0.5144	-0.5751	-0.4901	-0.4323	-0.7623	-0.7439	-0.7801	-0.7700	-0.7480	-0.7820
	ARI	-0.6693	-0.6854	-0.6191	-0.5172	-0.8152	-0.8071	-0.8537	-0.8180	-0.8090	-0.8570
	NMI	-0.5547	-0.5923	-0.6147	-0.6085	-0.7862	-0.8374	-0.8376	-0.7900	-0.8340	-0.8390
Spectral	Acc	-0.6883	-0.6145	-0.6112	-0.6031	-0.8169	-0.8257	-0.8602	-0.8200	-0.8280	-0.8580
	ARI	-0.6802	-0.6263	-0.6811	-0.6478	-0.8173	-0.8523	-0.8064	-0.8200	-0.8540	-0.8080
	NMI	-0.6184	-0.6391	-0.6519	-0.6102	-0.8185	-0.8841	-0.8762	-0.8200	-0.8820	-0.8730
Agglomerative	Acc	-0.4722	-0.5068	-0.6123	-0.4597	-0.7824	-0.7369	-0.7046	-0.7880	-0.7380	-0.7070
	ARI	-0.7439	-0.6874	-0.6012	-0.7325	-0.8647	-0.8099	-0.8865	-0.8660	-0.8130	-0.8890
	NMI	-0.8962	-0.7145	-0.7298	-0.6242	-0.8751	-0.9198	-0.8772	-0.8820	-0.9170	-0.8760
Deep Clustering	Acc	-0.4978	-0.4936	-0.6142	-0.5827	-0.8034	-0.8297	-0.8891	-0.8100	-0.8330	-0.8870
	ARI	-0.4781	-0.5279	-0.5831	-0.4852	-0.8457	-0.8192	-0.7931	-0.8480	-0.8200	-0.7980
	NMI	-0.5221	-0.5394	-0.5984	-0.6273	-0.8702	-0.7244	-0.7916	-0.8750	-0.7220	-0.7890

Table 12: Performance recovered by head-only adaptation after incremental backbone training on three generative models. JLT rows should be interpreted as recoverable performance under the diagnostic head-adaptation protocol, not as direct replacements for standard CIL accuracy. Joint Fine-tuning denotes fine-tuning on all tasks simultaneously. $\mathcal{A}_t$ is defined according to Equation 3, BWT according to Equation 1, and FWT according to Equation 2.

Model	**IL Method**	**Tacred**			**Banking77**			**Clinc150**			**Fewrel**			**Topic3**		
		$\mathcal{A}_t$	BWT	FWT	$\mathcal{A}_t$	BWT	FWT	$\mathcal{A}_t$	BWT	FWT	$\mathcal{A}_t$	BWT	FWT	$\mathcal{A}_t$	BWT	FWT
GPT2-NEOX (Pythia) 410M	Joint Fine-tuning	98.99	-	-	95.66	-	-	96.33	-	-	95.35	-	-	95.50	-	-
	SEQ*	44.34	0.00	0.00	67.12	0.00	0.00	84.51	0.00	0.00	61.99	0.00	0.00	70.56	0.00	0.00
	KLDA-E	97.20	0.00	0.00	93.03	0.00	0.00	96.62	0.00	0.00	94.55	0.00	0.00	94.53	0.00	0.00
	Base	11.97	-91.26	91.28	7.56	-75.42	68.86	4.67	-89.69	87.62	6.17	-80.57	73.59	20.53	-91.02	93.08
	L2KD	24.68	-79.27	94.39	52.32	-35.08	85.72	26.67	-73.17	94.74	30.18	-53.23	74.57	58.17	10.27	40.25
	LAMOL_KD	32.81	-69.55	93.68	49.48	-54.92	97.08	42.09	-60.48	97.50	27.70	-75.60	95.43	50.08	-54.98	92.97
	PCLL	24.30	-65.29	74.37	45.91	-56.97	84.41	44.16	-57.02	91.81	29.79	-68.41	82.84	55.80	-42.23	92.66
	Base + JLT	98.93	-0.55	11.38	75.07	-19.92	84.58	72.47	-25.50	92.29	51.31	-36.10	77.50	72.21	-14.80	79.92
	L2KD + JLT	98.04	-0.25	4.92	95.23	-2.99	88.56	96.20	-3.38	94.52	95.18	-1.04	93.52	93.91	-0.16	92.95
	LAMOL_KD + JLT	98.24	0.25	9.59	95.03	-3.33	88.98	96.38	-2.95	93.40	95.73	-0.54	92.82	94.00	-0.08	92.99
	PCLL + JLT	96.70	-0.58	10.29	93.31	-3.60	87.48	96.27	-3.29	91.36	94.70	-1.06	92.15	93.46	-0.26	93.51
LLaMA3.2 1B	Joint Fine-tuning	98.33	-	-	96.26	-	-	97.33	-	-	95.65	-	-	95.50	-	-
	SEQ*	45.82	0.00	0.00	67.48	0.00	0.00	83.72	0.00	0.00	62.13	0.00	0.00	71.64	0.00	0.00
	KLDA-E	97.36	0.00	0.00	93.27	0.00	0.00	96.71	0.00	0.00	94.68	0.00	0.00	94.59	0.00	0.00
	Base	12.23	-91.60	92.01	11.56	-91.10	89.28	5.89	-93.21	92.45	8.42	-81.39	77.03	19.50	-88.74	97.84
	L2KD	31.83	-68.77	81.20	51.56	-32.46	76.52	26.44	-66.62	87.88	32.75	-54.67	78.49	63.17	17.27	38.09
	LAMOL_KD	32.75	-62.04	86.04	45.85	-64.91	87.01	44.80	-54.05	92.93	29.96	-71.79	89.10	48.10	-56.19	92.72
	PCLL	31.18	-77.22	93.48	49.03	-67.69	87.40	40.69	-61.93	93.45	28.46	-74.38	93.88	48.84	-55.42	92.89
	Base + JLT	98.63	-0.74	21.56	79.47	-17.08	87.05	74.91	-23.33	93.45	51.29	-36.30	79.57	73.06	-13.23	79.85
	L2KD + JLT	97.06	-1.76	17.26	96.10	-2.20	87.46	96.36	-2.95	95.02	95.35	-1.00	94.38	93.82	-0.16	92.79
	LAMOL_KD + JLT	97.15	-0.83	8.44	95.36	-2.42	84.13	87.84	-8.10	89.05	94.72	-0.99	92.16	90.68	-1.03	89.74
	PCLL + JLT	96.30	-0.32	18.01	94.32	-2.54	87.33	96.29	-3.29	91.43	95.42	-0.97	94.94	92.24	-0.10	91.98
Qwen3 0.6B	Joint Fine-tuning	98.66	-	-	93.36	-	-	96.66	-	-	95.65	-	-	96.50	-	-
	SEQ*	44.09	0.00	0.00	66.84	0.00	0.00	82.91	0.00	0.00	60.77	0.00	0.00	70.41	0.00	0.00
	KLDA-E	96.94	0.00	0.00	92.81	0.00	0.00	96.25	0.00	0.00	94.33	0.00	0.00	94.21	0.00	0.00
	Base	11.30	-90.94	90.14	12.73	-92.39	91.44	6.60	-90.57	90.64	6.98	-79.11	73.01	19.99	-89.91	89.94
	L2KD	34.67	-67.41	90.38	55.78	-25.99	75.11	27.31	-63.83	86.02	32.90	-58.13	73.79	62.32	9.90	44.41
	LAMOL_KD	42.20	-59.61	94.17	52.44	-52.31	97.42	42.71	-59.83	93.07	26.45	-79.12	95.30	42.09	-60.48	93.50
	PCLL	49.40	-55.33	93.50	52.99	-50.68	97.78	43.27	-59.81	90.10	31.29	-74.50	96.14	49.40	-55.33	92.50
	Base + JLT	97.49	-1.73	15.76	75.26	-22.58	86.36	77.24	-21.31	93.24	52.55	-35.27	78.87	71.03	-18.32	82.20
	L2KD + JLT	98.21	-0.15	10.69	94.93	-3.67	86.40	95.27	-4.40	93.12	95.58	-1.15	94.98	92.16	-1.21	91.32
	LAMOL_KD + JLT	97.88	-0.53	13.95	92.08	-6.82	84.89	90.47	-9.50	92.07	95.01	-1.87	94.67	91.47	-1.53	90.21
	PCLL + JLT	96.23	-0.07	17.69	92.66	-5.00	87.93	95.22	-4.36	92.36	94.96	-1.13	96.55	89.56	-1.75	90.11

E.3 Model size & capacity

In Figure 11, we measured BWT-backbone, FWT-backbone, and their sum using the Base method in three groups of representative LLMs (GPT-style Pythia, Llama, and Qwen). The sum $BWT + FWT = a_{T,i} - a_{i-1,i}$ cancels the shared post-training term $a_{i,i}$, so we treat it as a coupling-aware net-transfer view rather than as a primary metric. The value increases with model size within the same model series. In particular, as model size increases, BWT-backbone improves, indicating a positive relationship between model size and retention. On the other hand, FWT-backbone remains nearly constant across models and sizes. The relatively stable operational FWT despite larger model sizes should be considered when designing future IL methods.



(a) Pythia (b) Llama3 (c) Qwen3

Figure 11: BWT-backbone, FWT-backbone, and BWT-backbone + FWT-backbone results in Pythia, Llama, and Qwen, measured over five benchmark groups with the Base method.

Table 13: Hyperparameters for LM-head adaptation.

Hyperparameter	Default
lm_head_lr	e.g. 5e-4
Optimizer	AdamW
Loss (LM)	Cross-entropy
training_epoch	3
Batching	Same as backbone training loop

F LM-head adaptation details

Setup. For the JLT diagnostic, the backbone is fixed after sequential IL training and only the LM head is updated. We use two head-adaptation modes. In post-hoc JLT, the head is trained once after the full task sequence using the data allowed by the IL setting. In task-adaptive JLT, the backbone is frozen and the LM head is briefly adapted on the training split of the current task before that task is evaluated (see Section F.1). The latter is an oracle-style recovery diagnostic.

Label tokenization. For each instance, we tokenize the gold textual label and take the first token id as target:

$$y^{\text{tok}} = \text{Tokenizer}(\text{label})[0].$$

Given hidden feature $\mathbf{h} \in \mathbb{R}^d$ from the backbone and output embedding $W \in \mathbb{R}^{V \times d}$, the vocabulary logits are

$$\mathbf{z} = W\,\mathbf{h} \in \mathbb{R}^V.$$

Base LM-head loss.

$$\mathcal{L}_{\text{LM}} = \text{CE}\big(\mathbf{z},\, y^{\text{tok}}\big)\,.$$

Equivalently, for task n with N_n examples, hidden features $h_n^i = f_\theta(x_n^i)$, and label-token mapping $\tau(y)$, the task-level head-only objective is

$$\mathcal{L}_n = -\frac{1}{N_n}\sum_{i=1}^{N_n} \log \text{softmax}(W h_n^i)\big[\tau(y_n^i)\big].$$

Optimization. The LM head parameters are optimized by AdamW with learning rate η (default chosen as $5\text{e}{-4}$ in our runs). Backbone and optional external classifiers are trained with a separate optimizer.

F.1 Evaluation-time LM-head finetune

Before evaluation on task t, the backbone is frozen and the LM head is adapted for E epochs (typically 1–3) on the training split of task t using $\mathcal{L}_{\text{LM}}$. After evaluation, LM-head weights are restored to their original state to avoid leakage across tasks. Because this protocol uses task-specific training data at evaluation time, it is reported only as a diagnostic estimate of recoverable head performance.

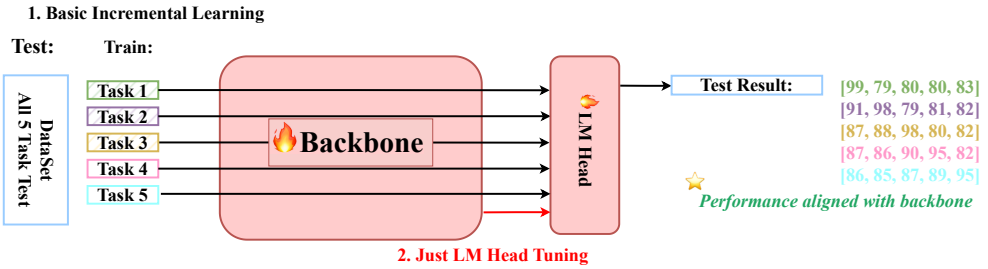


Figure 12: Overview of JLT as a head-only diagnostic. The IL process follows the basic procedure of each IL method. After IL training, the backbone is fixed and the LM head is adapted to estimate how much information in the backbone can be recovered through head re-alignment.

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